

Symmetric Group Practice 1

1. Let α be the 5-cycle $(1\ 2\ 3\ 4\ 5)$.

a) Write each distinct power of α in cycle notation.

$$\alpha^1 = (1\ 2\ 3\ 4\ 5) \quad \alpha^2 = \quad \alpha^3 =$$

b) How many distinct powers do you get for a 5-cycle?

c) What is the connection between the powers of α and modular arithmetic?

d) What is α^{-1} ?

For what positive power n is $\alpha^{-1} = \alpha^n$?

e) Which exponent of α would you use to represent the identity permutation?

Definition: If x is any element of a group G , the order of x is the smallest positive integer n such that x^n is the identity in the group.

2. What is the order of the permutation $(1\ 2)(3\ 4\ 5)$?

3. What is the order of the permutation $(1\ 2)(3\ 4\ 5\ 6)$?

4. What is the order of the permutation $(1\ 2)(3\ 4\ 5)(6\ 7\ 8\ 9)$?

5. Find two elements of order 15 in S_9 .

6. What is the largest possible order for an element in S_9 ? S_{10} ? S_{11} ?

7. Let $\alpha = (1\ 2\ 3\ 4)$ and $\beta = (1\ 2\ 3)$ be permutations in S_4 (so β leaves 4 fixed).

a) Write each permutation in array form.

$\alpha =$

$\beta =$

b) Write the product $\alpha\beta$ in both array form and in cycle form.

c) What is the order of α ?

What is the order of β ?

d) What is the order of the product $\alpha\beta$?

8. Fill in the Cayley table for S_3 from our notes in class.

$\circ$	ϵ	α	β	γ	δ	φ
ϵ						
α						
β						
γ						
δ						
φ						

$$\epsilon = (1)$$

$$\alpha = (2\ 3)$$

$$\beta = (1\ 2)$$

$$\gamma = (1\ 2\ 3)$$

$$\delta = (1\ 3\ 2)$$

$$\varphi = (1\ 3)$$